THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **The Wonderful World of Science, Diversity in the Living World, Mindful Eating: A Path to a Healthy Body, Exploring Magnets** | Class 6 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** the volume of water is changed along with the time, so the test has two changing things	Volume of water is a variable.
2	**(a)** a grass	Grasses show these features.
3	**(d)** starch and fat	Both tests are positive.
4	**(a)** P and R are magnets, Q is magnetic material	Repulsion is seen only between magnets.
5	**(a)** both depend on hot weather	A third factor links them.
6	**(c)** helps judge distance for hunting	Forward eyes give depth perception.
7	**(b)** they contain much sugar and few nutrients	Empty calories.
8	**(a)** iron is magnetic and copper is not	Only iron is attracted.
9	**(d)** compare the methods and repeat carefully	Differences in method may explain.
10	**(c)** many plants will fail to pollinate	Pollination would fall.
11	**(b)** protein	Protein test.
12	**(b)** south pole	Opposite pole forms at the lifting end.
13	**(d)** more than one variable was changed	Only one variable should vary.
14	**(b)** amphibian	Frogs show this.
15	**(a)** Q contains starch	Iodine tests starch only.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Two factors differ: light and water (1); We cannot tell which factor caused the difference (1).	1+1
17	Green stem carries out photosynthesis (1); Spines reduce water loss (1).	1+1
18	It contains starch (1); It also contains fat (1).	1+1
19	The steel table disturbs the needle (1); Move it to a wooden or non-magnetic surface (1).	1+1
20	It represents the Earth in small size (1); It helps us study things too large to observe wholly (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Use same size, same wetness (1); Same place, time and temperature (1); Measure drying time for each (1).	1+1+1
22	Tree (1); Tap root goes deep (1); Net veins in leaves (1).	1+1+1
23	Lacks protein and vitamins (1); High in refined carbohydrates and fat (1); Add milk, dal, vegetables and fruit (1).	1+1+1
24	Roll bar in filings (1); Maximum cling near the ends (1); Little in the middle (1).	1+1+1
	OR: Curiosity: why do things fall (1); Observation: fall of apple and Moon's motion (1); Reasoning: same force acts on both (1).	1+1+1
25	Colour can vary within a group (1); Unrelated animals may have the same colour (1); Features like body covering and habitat are better (1).	1+1+1
	OR: Calcium for bones (1); Proteins for growth (1); Vitamins like A and D (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Hypothesis: dark cups cool faster (1); Same volume and temperature of water (1); Cups of different colour, same size and material (1); Measure temperature every 5 minutes (1); Plot results and conclude (1).	1+1+1+1+1
	OR: Aquatic: fish, whale (1); Terrestrial: cow, lion (1); Aerial: crow, eagle (1); Herbivore/carnivore/omnivore assigned (1); Features such as fins, legs, wings explained (1).	1+1+1+1+1

27	A: vitamin C, citrus fruits (1); B: vitamin A, carrot (1); C: iodine, iodised salt (1); D: vitamin D, sunlight and milk (1); E: iron, spinach (1).	1+1+1+1+1
	OR: Same type of clips (1); Same method to approach (1); Count clips lifted by each pole (1); Repeat three times (1); Compare averages (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **The Wonderful World of Science, Diversity in the Living World, Mindful Eating: A Path to a Healthy Body, Exploring Magnets** | Class 6 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** south-seeking pole	The end that points south is the south-seeking pole.
2	**(b)** amount of water	Water amount varies.
3	**(b)** wide field of view to spot predators	Wide view.
4	**(b)** protein and fat	Plant foods have less protein and fat.
5	**(a)** iron only	Only iron is magnetic.
6	**(a)** rounding off numbers to look neat without noting it	Data must not be altered.
7	**(b)** it provides more food, water and shelter	Rich resources support life.
8	**(a)** X starch, Y fat, Z protein	Match tests with nutrients.
9	**(d)** B's north repels A's north	Like poles repel.
10	**(b)** revise the hypothesis	Data must be respected.
11	**(a)** to avoid sinking in sand	Large area reduces pressure.
12	**(c)** spinach and jaggery	Iron-rich foods.
13	**(b)** repel	N facing N.
14	**(a)** growth of the plant	Growth is the dependent variable.
15	**(c)** being active at night	Avoid heat.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	It breathes air through lungs (1); It gives birth to young and feeds milk (1).	1+1
17	Lacks carbohydrates and fibre (1); Lacks many vitamins (1).	1+1
18	No (1); Heat weakens the magnet (1).	1+1
19	Example: plastic waste (1); Science develops recycling or biodegradable materials (1).	1+1
20	To take in air (1); Mud around them has little oxygen (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Permanent magnet is present (1); A coil moves in its field (1); Produces sound (1).	1+1+1
22	A factor that can change (1); Temperature of water (1); Amount of sugar and stirring (1).	1+1+1
23	Beak shape matches food (1); Woodpecker has strong beak for wood (1); Hummingbird has thin beak for nectar (1).	1+1+1
24	Fruits and nuts (1); Roasted chana (1); Avoid sugary packaged items (1).	1+1+1
	OR: Region around a magnet (1); Where force can be felt (1); Shown by iron filings (1).	1+1+1
25	P vs Q tests light (1); P vs R tests water (1); Only one factor differs in each comparison (1).	1+1+1
	OR: Duck: webbed feet (1); Fish: fins and tail (1); Fish: streamlined; duck: oily feathers (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Two sets of same plants (1); One with music, one without (control) (1); Same water, soil, light (1); Measure height weekly (1); Compare and conclude (1).	1+1+1+1+1
	OR: Animals lose food and shelter (1); Plant cover declines (1); Plant more trees (1); Create national parks (1); Avoid pollution and hunting (1).	1+1+1+1+1
27	Cereal for energy (1); Dal for proteins (1); Vegetables for vitamins (1); Curd for calcium (1); Salad and pickle for fibre and flavour (1).	1+1+1+1+1

	OR: Roll magnet in filings (1); Filings gather at the ends (1); Fewer in middle (1); Count or compare quantity (1); Poles are strongest (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **The Wonderful World of Science, Diversity in the Living World, Mindful Eating: A Path to a Healthy Body, Exploring Magnets** | Class 6 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** cereals with pulses	Combination improves protein quality.
2	**(d)** Earth's field aligns the tiny magnets	Induction by Earth's field.
3	**(a)** Why do wet clothes dry faster on a windy day?	It looks for a cause.
4	**(b)** animals use oxygen produced by plants	Plants give oxygen.
5	**(c)** easily digestible, nutritious food	Recovery needs nutrients.
6	**(a)** unlike poles attract	Earth's north is magnetic south.
7	**(c)** Does temperature affect how fast sugar dissolves?	Testable questions have measurable answers.
8	**(b)** thick fur, fat layer, white colour	Cold adaptation.
9	**(c)** vitamin A, night blindness	Vitamin A deficiency affects dim-light vision.
10	**(a)** a magnet or iron object nearby	Local magnetic influence.
11	**(c)** careful observation matters more than costly equipment alone	Simple tools with careful thinking can lead to discoveries.
12	**(c)** prop roots	They support the tree.
13	**(a)** butter and dal	Butter has fat; dal protein.
14	**(c)** the pin acts as a temporary magnet	Induced magnetism.
15	**(b)** the plant grows with time	Rising line indicates growth.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	For growth of tissues (1); For repair of body cells (1).	1+1
17	A pin can be moved through paper by a magnet (1); Non-magnetic sheets do not block force (1).	1+1
18	It shows what happens without the change (1); Example: unfertilised plants compared to fertilised ones (1).	1+1
19	No (1); Bats and insects also fly (1).	1+1
20	Little sunlight reduces vitamin D making (1); They can use milk, eggs or supplements (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Dal adds protein (1); Dal adds minerals (1); Together they give a more balanced intake (1).	1+1+1
22	Gave direction at sea (1); Not dependent on stars (1); Safer voyages (1).	1+1+1
23	Hand is unreliable (1); It depends on earlier experience (1); A thermometer gives exact readings (1).	1+1+1
24	Stores fat (1); Fat gives energy when food is scarce (1); Body fat in one place keeps the rest cooler (1).	1+1+1
	OR: Vitamins get destroyed by air and light (1); Wash before cutting (1); Cut just before eating (1).	1+1+1
25	Hang a bar magnet freely (1); It settles N-S (1); Repeated turns give same result (1).	1+1+1
	OR: New experiments contradict it (1); A better explanation covers more facts (1); Example: Earth at centre idea was given up (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Sum = 90 s (1); Average = 30 s (1); One swing = 1.5 s (1); Timing many swings reduces error in starting and stopping (1); Dividing gives the time for one swing (1).	1+1+1+1+1
	OR: Hollow bones: lightness (1); Feathers: flight and warmth (1); Beak: food type (1); Feet: perching, swimming or hunting (1); Examples: duck, eagle, parrot (1).	1+1+1+1+1

27	Breakfast: milk, fruit, paratha (1); Lunch: roti, dal, vegetable, curd (1); Snack: sprouts or fruit (1); Dinner: light khichdi with vegetables (1); Avoid chips, soda and excess sweets (1).	1+1+1+1+1
	OR: Test each pair for attraction (1); Copper is attracted by none (1); The magnet may repel another bar's end (1); Iron is attracted by the magnet but not repelled (1); All three identified (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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